

CT of the Difficult Acute Aortic Syndrome

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Abbreviations: AAS = acute aortic syndrome, AD = aortic dissection, EKG = electrocardiography, IMH = intramural hematoma, PAU = penetrating atherosclerotic ulcer, PE = pulmonary embolism, SVS = Society for Vascular Surgery, TEE = transesophageal echocardiography, ULP = ulcerlike projection

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SA-CME LEARNING OBJECTIVES

After completing this journal-based SA-CME activity, participants will be able to:

- Understand the multidisciplinary American Heart Association classifications of intimal tears.
- Identify flow artifacts and issues regarding phase of contrast enhancement that may complicate diagnosis of AAS.
- Recognize unusual signs of aortic rupture and its mimics.

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Acute aortic syndrome (AAS) is classically attributed to three underlying pathologic conditions—aortic dissection (AD), intramural hematoma (IMH), and penetrating atherosclerotic ulcer (PAU). In the majority of cases, the basics of image interpretation are not difficult and have been extensively reviewed in the literature. In this article, the authors extend existing imaging overviews of AAS by highlighting additional factors related to the diagnosis, classification, and characterization of difficult AAS cases. It has been well documented that AAS is caused not only by an AD but by a spectrum of lesions that often have overlap in imaging features and are not clearly distinguishable. Specifically, phase of contrast enhancement, flow artifacts, and flapless AD equivalents can complicate diagnosis and are discussed. While the A/B dichotomy of the Stanford system is still used, the authors subsequently emphasize the Society for Vascular Surgery's new guidelines for the description of acute aortic pathologic conditions given the expanded use of endovascular techniques used in aortic repair. In the final section, atypical aortic rupture and pitfalls are described. As examples of pericardial and shared sheath rupture become more prevalent in the literature, it is important to recognize contrast material third-spacing and mediastinal blood as potential mimics. By understanding these factors related to difficult cases of AAS, the diagnostic radiologist will be able to accurately refine CT interpretation and thus provide information that is best suited to directing management.

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Introduction

To facilitate early clinical suspicion for nontraumatic acute aortic pathologic conditions, the term *acute aortic syndrome* (AAS) was introduced in 2001 to describe a group of patients with hypertension who presented with similar intense, acute, throbbing, and/or migratory chest or back pain—so-called aortic pain. In this original description, AAS was attributed to three underlying causes: aortic dissection (AD), intramural hematoma (IMH), or penetrating atherosclerotic ulcer (PAU) (1). The term AAS and its association with these conditions has propagated, and it is now often used to describe the group of pathologic conditions themselves rather than the clinical syndrome (2–4).

TEACHING POINTS

- CT features that can implicate an isolated PAU as the cause of acute symptoms include interval development of the lesion from a recent prior examination, progression of the lesion at short-term follow-up, large size, deep ulcer depth, and signs of impending or active rupture (high-attenuating blood in an associated thrombus, adjacent stranding or blood products outside the PAU).
- Given the subtlety of the focal bulge on CT images, other findings should trigger careful scrutiny of the aorta and increase suspicion for a class III lesion. The first is the presence of signs of aortic rupture without an immediately identifiable class I or class II lesion. Class III lesions are more common in the ascending aorta and can be associated with blood in the mediastinum, hemopericardium, and rupture into the shared adventitial sheath with the main pulmonary artery. Class III lesions also often result in focal dilatation of the aorta or may arise in patients with ascending aortic aneurysms.
- There are several features of flow artifacts that can assist in distinguishing them from a true intimomedial flap. Given that they are caused by mixing of opacified and nonopacified blood, flow artifacts typically have higher attenuation than an intimomedial flap. More importantly, flow artifacts often have indistinct borders and discontinuities in at least one plane and have been described as “smoke-like.”
- While the A/B dichotomy of the Stanford system is still used, a more detailed anatomic description of the extent of acute aortic pathologic conditions using the aortic zones as originally described by Criado and Ishimaru is encouraged.
- A potential pitfall in the detection of pericardial and pleural rupture is third spacing of contrast material from previous CT examinations. In patients who have undergone recent CT examinations, contrast material can accumulate in nonbloody pleural and pericardial effusions and simulate the presence of blood products.

As a result of its wide availability, rapid acquisition, and excellent diagnostic performance, multidetector CT has become the primary means of evaluation for suspected noncardiac causes of chest pain in patients presenting to the emergency department. In terms of AD, CT has published sensitivities and specificities near 100% (5). Despite this data, a recent study of patients with Medicare claims determined that 4.5% of ADs were missed in the emergency department (6,7). While some patients may have presented with a clinical picture that did not warrant imaging at initial presentation, others did not receive an accurate diagnosis despite prompt acquisition of a CT examination. Indeed, missed acute aortic pathologic conditions at CT have been observed in our own practice, both in patients initially presenting to our institution and in patients transferred to our quaternary care center. In addition to the rare completely missed diagnoses, cases of nearly missed or poorly described acute aortic pathologic conditions have also been encountered. To make matters more concerning, we have observed that the majority of these difficult diagno-

ses involve the ascending aorta, in which a missed or delayed diagnosis is more likely to result in a deleterious outcome.

Many authors have provided excellent overviews of imaging of AAS, including recent publications by Ko et al (8) and Murillo et al (9). The purpose of this article is to extend this work and highlight additional issues encountered in difficult AAS cases. Broadly, these are divided into factors related to diagnosis, classification, and rupture. For diagnosis, important concepts described include flapless AD equivalents, issues with phase of contrast enhancement, and flow artifacts. Subsequently, issues with classification and the need for precise communication are addressed. In the final section, difficulties in characterizing aortic rupture are reviewed on the basis of our own experience with AAS. We hope that this article will accelerate the reader's practice and refine CT interpretation of difficult AAS cases.

The Difficult AAS Diagnosis

Fundamentally, acute aortic pathologic conditions involve disruption of the aortic media. Most often, this is due to an intimomedial tear that occurs on the luminal side of the aortic wall (10,11). When blood has access to the media, a false lumen, separated by an intimomedial flap, can be created. This is the pathophysiologic basis for AD, the most common acute aortic pathologic condition, accounting for between 70% and 95% of cases of AAS (12–14). The best available data have demonstrated that CT is 98%–100% sensitive and 98% specific for the diagnosis of AD (5). CT is particularly well suited to help in the diagnosis of AD as it can help identify the false lumen, the intimomedial flap, and the entry tear, which is the most proximal site of communication between the true and false lumen.

As has been well documented, however, AAS is caused not only by AD but rather by a spectrum of lesions (6,8,9,15,16). The various types of intimal tears that can cause aortic pathologic conditions were further subclassified by Svensson and later adopted into the multispecialty American Heart Association guidelines in 2010 (15,16) (Table). Class I–IV intimomedial defects arise in the absence of a known external force, while class V lesions are due to iatrogenic (typically vascular catheterization) or traumatic causes. Class I intimomedial defects describe classic AD and result in the “double-barreled” aortic lumen with true and false lumina separated by an intimomedial flap. Class II lesions, known as IMHs, are characterized by hematoma in the aortic wall without a detectable tear or intraluminal flap (13,15). By virtue of improved preoperative imaging and the results of autopsy studies, it is now known that

Svensson Classification of Intimal Tears

Class	Description	Key CT Findings
I	Classic dissection with intimal tear resulting in a dissection flap	Intimomedial flap separating true and false lumen
II	IMH	Hyperattenuating crescent on a noncontrast image No substantial narrowing of the aortic lumen Focal contrast enhancement within the wall, representing an ulcerlike projection (ULP) or intramural blood pool
III	Intimal tear without medial hematoma	Subtle irregularity of the aortic wall Small focal defects or outpouching Strands of intimomedial or medial tissue extending into the lumen Lack of atherosclerosis Subadventitial blood tracking Signs of rupture
IV	PAU	Focal projection of contrast material beyond the intima Localized enlargement of the aorta or saccular pseudoaneurysm Extensive atherosclerosis Most commonly located in mid-distal descending thoracic aorta
V	Iatrogenic and traumatic dissections	Tear begins at site of coronary artery ostium May be associated with coronary thrombosis

the majority of patients with IMH have associated often small intimomedial defects; this has resulted in some authors considering IMHs as thrombosed variants of ADs (13,17–19).

Whether due to thrombosis of an AD or not, IMH represents a distinct CT condition from classic AD in that there is no discernible flowing blood within the false lumen (20). The imaging findings of IMH are relatively straightforward but may be subtle and include hyperattenuating typically crescentic thickening of the aortic wall that is often most easily appreciated on noncontrast acquisitions (8–10). While IMH is a well-recognized cause of AAS that will certainly be encountered by the radiologist in practice, the performance of CT in the diagnosis of IMH is not known. This is in part owing to its relatively decreased frequency in comparison with classic AD, as well as the overlapping and often coexisting CT appearance of the two conditions (21). It should be known that in 2021, with the advent of modern CT scanners, radiologists have the best in vivo view of acute aortic lesions, and the final determination of whether a lesion is in class I or class II may not be satisfactorily resolved either in the operating room or with pathology specimens (22,23). As such, essentially all radiology publications on IMH have focused on its imaging features with the purported goal of helping radiologists not miss these lesions. This is of particular importance given that IMH is typically managed in the same fashion as AD, with lesions involving the ascending aorta managed surgically, while lesions involving the aorta distal to the brachiocephalic artery are managed with combinations of medical, surgical, and endovas-

cular techniques dependent on patient-specific factors and the presence of complications (24).

Class IV lesions, otherwise known as PAUs, involve the ulceration of an atherosclerotic plaque into the media. This can result in a local pseudoaneurysm or longitudinal propagation in the form of an AD or IMH (10). While PAUs are lumped together with AD and IMH as causes of AAS, it is important to recognize that the overwhelming majority of PAUs do not result in AAS or clinical symptoms (12). Given that most patients with PAU are asymptomatic, it is often impossible to know whether an identified PAU is the cause of a patient's symptoms. While PAUs associated with AD or IMH are managed more aggressively and on the basis of the classification of the AD or IMH, the decision of whether to treat isolated class IV lesions in patients with suspected AAS often becomes controversial in real practice, with CT features potentially playing a role in diagnosis (25). CT features that can implicate an isolated PAU as the cause of acute symptoms include interval development of the lesion from a recent prior examination, progression of the lesion at short-term follow-up, large size, deep ulcer depth, and signs of impending or active rupture (high-attenuating blood in an associated thrombus, adjacent stranding or blood products outside the PAU) (26,27) (Fig E1). Therefore, in practice, the complexities of PAU in the setting of AAS are not related to recognizing the lesion itself, which is well described and easily understood, but rather in determining the clinical relevance of the lesion when it is encountered.

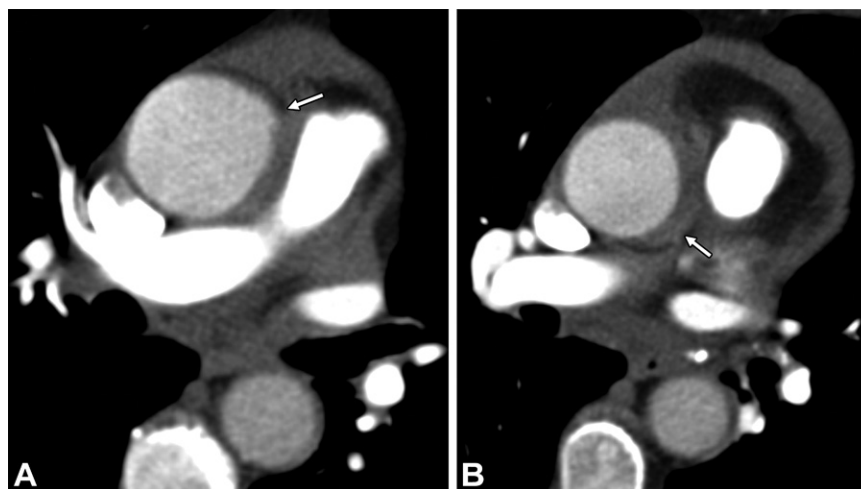


Figure 1. Missed limited intimal tear (Svensson class III lesion) in a 70-year-old woman who presented to an outside hospital with chest pain and was transferred to our institution with a diagnosis of pericardial effusion. **(A)** Axial CT image shows subtle irregularity and bulging along the medial aspect of the proximal ascending aorta (arrow). **(B)** Axial CT image shows a crescent of blood (arrow) tracking inferiorly along the medial wall of the aorta with hemopericardium. While these images were being reviewed, the patient became unstable and died in the emergency department. The results of an autopsy demonstrated a limited intimal tear with subadventitial and pericardial rupture.

The Flapless Acute Aorta (Limited Intimal Tear)

Class III lesions, described as intimal tears without mural hematoma, were originally recognized in the 1970s (28). Largely forgotten in the next several decades, Svensson rekindled interest in these lesions in 1999 with his description of nine patients who each had an acute aortic pathologic condition identified at surgery that was not detected at preoperative imaging (16). More recently, Chin et al (29) retrospectively identified 24 patients with similar lesions, identifying them as 4.8% of 497 patients presenting with AAS at a tertiary care institution. Of note, four of the 24 patients in this series did not have the diagnosis made at preoperative imaging (29). This has mirrored our experience, as we have seen class III lesions missed at our own institution.

At surgery, class III lesions have been described as stellate or linear intimal tears with exposure of the underlying media or adventitia but without propagation of a false lumen or hematoma in the wall (30). Several names have been used to describe these uncommon lesions, including “incomplete dissections” and “limited intimal tears (LITs)” (16,29). CT features of class III lesions have focused on the identification of a focal bulge or contour defect in the aortic wall. This focal bulge may be difficult to identify on transaxial images, sometimes completely obscured owing to the curvature of the aorta, mistaken for pulsation artifact, or appearing as a focal flaplike luminal defect when one of the edges is viewed en face. Multiplanar reconstructions

are of value in depicting the focal bulge often visible in these cases (8,29). This focal contour bulge should not be associated with substantial atherosclerosis. The lack of atherosclerosis is a CT feature that allows distinguishing of class III from class IV lesions, which are far more common and typically manifest in older patients in the descending aorta.

Given the subtlety of the focal bulge on CT images, other findings should trigger careful scrutiny of the aorta and increase suspicion for a class III lesion. The first is the presence of signs of aortic rupture without an immediately identifiable class I or class II lesion (29) (Fig 1). Class III lesions are more common in the ascending aorta and can be associated with blood in the mediastinum, hemopericardium, and rupture into the shared adventitial sheath with the main pulmonary artery. Class III lesions also often result in focal dilatation of the aorta or may arise in patients with ascending aortic aneurysms. When acute, the focal aortic dilatation associated with these lesions can result in aortic insufficiency due to failure of coaptation of the valve leaflets. In addition to the focal bulge, the underlying exposed media may produce a wavy or irregular contour to the inner aortic wall (29). Indeed, we have observed this wavy or irregular inner wall in cases in which the focal contour bulge is identified, often extending beyond the focal contour bulge and defining the full extent of the lesion, which is easy to underestimate and was reported to be up to 7.7 cm in the series reported by Chen et al (29) (Fig 2).



Figure 2. Long segment limited intimal tear (Svensson class III lesion) in a 43-year-old man who presented with chest pain. (A) Oblique coronal CT image shows a focal bulge with an overhanging edge (arrow) along the lesser curvature of the distal ascending aorta. Extending distally, there is some irregularity in the aortic wall, consistent with exposed media (arrowheads). (B) Axial CT image at the level of the aortic arch shows tracking blood along the arch with a focal contour bulge (arrow) depicted laterally at the level of the left subclavian artery. At surgery, a long segment intimomedial tear was found from the bulge in the distal ascending aorta to the bulge in the lateral arch. This lesion required ascending aorta and arch repair.

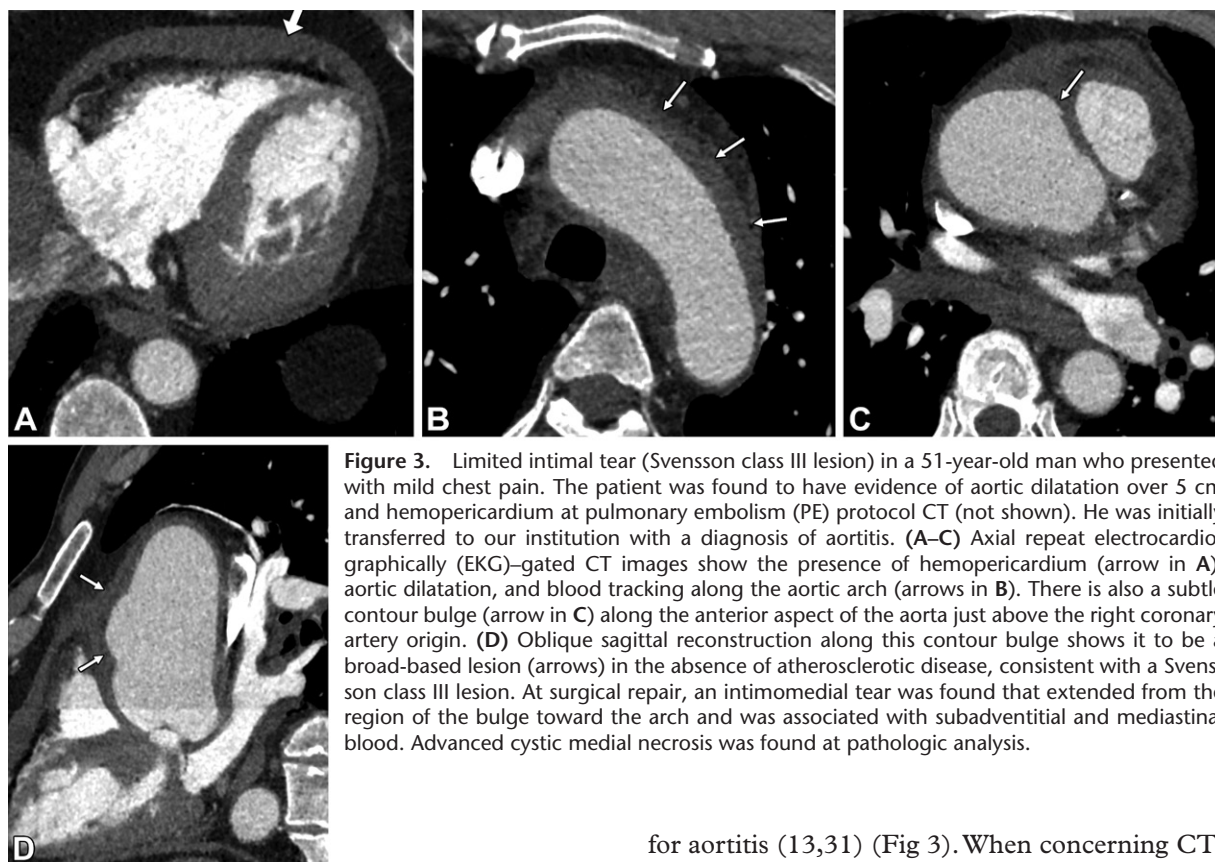


Figure 3. Limited intimal tear (Svensson class III lesion) in a 51-year-old man who presented with mild chest pain. The patient was found to have evidence of aortic dilatation over 5 cm and hemopericardium at pulmonary embolism (PE) protocol CT (not shown). He was initially transferred to our institution with a diagnosis of aortitis. (A–C) Axial repeat electrocardiographically (EKG)-gated CT images show the presence of hemopericardium (arrow in A), aortic dilatation, and blood tracking along the aortic arch (arrows in B). There is also a subtle contour bulge (arrow in C) along the anterior aspect of the aorta just above the right coronary artery origin. (D) Oblique sagittal reconstruction along this contour bulge shows it to be a broad-based lesion (arrows) in the absence of atherosclerotic disease, consistent with a Svensson class III lesion. At surgical repair, an intimomedial tear was found that extended from the region of the bulge toward the arch and was associated with subadventitial and mediastinal blood. Advanced cystic medial necrosis was found at pathologic analysis.

Finally, the focal bulge can manifest with blood tracking along or in the wall of the aorta in a longitudinal fashion. This blood may be subadventitial and can simulate IMH or be confused

for aortitis (13,31) (Fig 3). When concerning CT findings are depicted on nongated images, a repeat acquisition with electrocardiographic (EKG) gating can be of value in better defining them and increasing diagnostic confidence.

Patient characteristics can also increase suspicion. Class III lesions are frequently seen

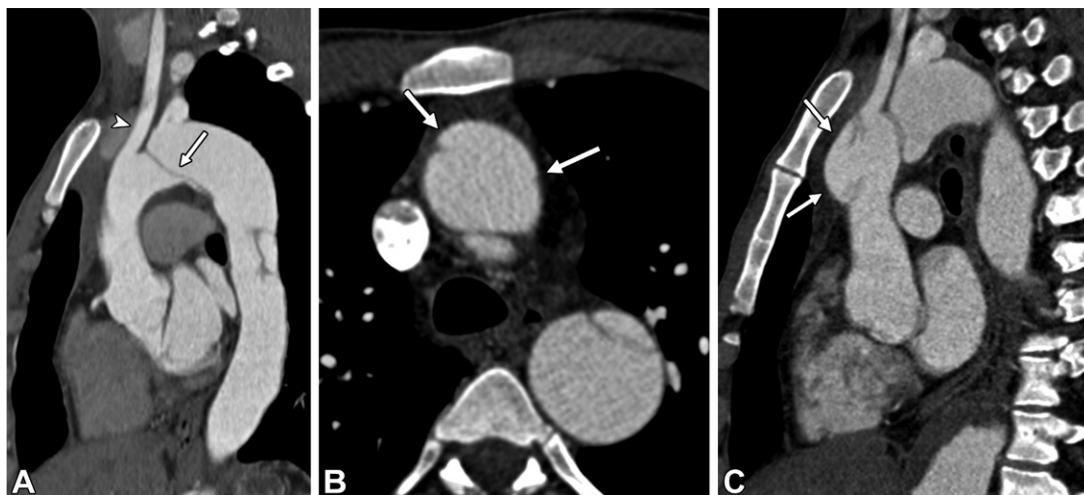


Figure 4. Limited intimal tear (Svensson class III lesion) with pseudoaneurysm formation in a 22-year-old man with chest pain and underlying aortopathy (ACTA2 mutation). (A) Sagittal oblique CT image shows the intimal flap (arrow) terminating just distal to the innominate artery (arrowhead) in the arch. This dissection was managed medically. Five years later, the patient presented again with chest pain. (B) Axial CT image shows a new broad-based bulge (arrows) with overhanging edges and small adjacent blood along the anterior aspect of the ascending aorta. (C) Oblique sagittal reconstruction CT image confirms the extent of the pseudoaneurysm and shows the overhanging edges (arrows) to advantage. Open surgical repair confirmed the presence of the intimal defect with pseudoaneurysm and adjacent rupture in this young patient with genetic aortopathy and no atherosclerotic disease.



Figure 5. Ruptured limited intimal tear (Svensson class III lesion) in a 67-year-old man who presented to the emergency department with vague brief chest pain that spontaneously resolved. Twenty-four hours after initial presentation, CT was performed. (A) Axial CT image shows irregularity along the medial wall of the aorta (arrow) consistent with exposed media. (B) Oblique sagittal reconstruction at this level correlates with a focal bulge in the aortic wall (arrow) consistent with a limited intimal tear (Svensson class III lesion). Of note, the fat adjacent to the bulge was normal. (C) Additional oblique sagittal reconstruction shows blood within the mediastinum. In the proximal descending aorta, there was another large pseudoaneurysm with some internal thrombus but no adjacent atherosclerosis. This lesion was subsequently found to be present at examinations performed 6 months earlier and was chronic. At surgery, aortic rupture was found in association with the limited intimal tear in the ascending aorta. A two-stage hybrid repair of the ascending aorta, aortic arch, and proximal descending aorta was performed. Pathologic analysis results confirmed advanced cystic medial necrosis.

in younger patients without substantial atherosclerotic disease and are frequently associated with conditions that result in early and advanced medial degeneration, including connective tissue diseases, bicuspid aortic valve, and hereditary aortopathies (29) (Fig 4). In addition, in these patients with early and advanced medial degeneration, there may be multiple lesions of different acuity levels caused by intimal medial tears (Fig 5). In our experience, clinical symptoms,

including chest pain, may be vague or absent in these patients. Given their complex clinical and CT presentation, identification of Class III lesions at CT requires a high index of suspicion, careful multiplanar reconstructions, and the use of multiple findings to make the diagnosis. As such, when suspected, we describe the imaging features and often communicate them as flapless AD equivalents (Svensson class III variant) to fully implicate the importance of the findings.

This has often been necessary in these difficult cases, as even seasoned surgeons may be reticent to operate without definitive signs of an intimo-medial flap or IMH.

The Importance of Phase of Contrast Enhancement

At dissection protocol CT, which most commonly involves a noncontrast acquisition followed by a postcontrast acquisition timed in the systemic arterial phase, identification of class I, II, IV, and V lesions is almost always straightforward. This is not to imply that subtle examples of these pathologic conditions may not be encountered at CT, but rather that the findings even in subtle cases are typically identifiable prospectively when interpreted by experienced readers. Unfortunately, patients with class I, II, IV, and V lesions can have variable clinical manifestations. A review of the International Registry of Aortic Dissections sheds more light on the variable presentations of patients with acute aortic pathologic conditions, with substantial numbers of patients reporting no pain or denying symptoms typical of aortic pain (32). In addition, the majority of patients with acute ascending aortic pathologic conditions were not hypertensive at presentation (32). Thus, the clinical presentation of many patients with aortic pathologic conditions, particularly those involving the ascending aorta, will overlap with those of other conditions, including acute coronary syndrome, pulmonary embolism (PE), and pneumothorax.

These clinical factors can result in patients undergoing CT examinations that are not AD protocols, including PE protocol CT. PE protocol CT examinations are single-phase examinations in which opacification of the systemic arteries including the aorta does not reliably occur. At PE protocol CT examinations obtained in a phase prior to complete aortic opacification with contrast material, suspicion for an aortic cause of the patient's symptoms often relies on detection of aortic enlargement, focal contour defects in the wall of the aorta, or findings suggestive of potential aortic rupture (33,34). In addition, routine use of narrow windows should be performed to evaluate for intimo-medial flaps that may be detectable even with no or weak opacification of the aorta (Fig E2). When concerning findings are identified at PE CT, options for definitive characterization include repeating the CT examination with a standard dissection protocol or, in patients in whom the additional use of iodinated contrast material is potentially problematic, performing transesophageal echocardiography (TEE) or MRI for further evaluation.

In some cases, there may be opacification of the aorta, but it will be in an early systemic arte-

rial phase. This may not only occur at PE CT but also at AD CT protocol examinations on faster CT scanners or those that are performed in patients with decreased cardiac function. In the early systemic arterial phase, there will be limited organ enhancement and no contrast present in the portal vein. This has important implications for the detection of acute aortic pathologic conditions. In a classic AD, the false lumen will almost always enhance after the true lumen; therefore, in the systemic arterial phase, the false lumen may not be enhancing, may be weakly enhancing, or may have a gradient of enhancement extending from the entry tear (Fig 6). In some cases, this gradient of enhancement in the false lumen can be used to advantage in identifying the entry tear, which can be seen as the point where there is equilibration of enhancement of the true and false lumens.

If the false lumen is thin, it may have low attenuation and thus be difficult to identify in the systemic arterial phase, in some cases potentially confused with a pericardial recess, fluid adjacent to the aorta, or atherosclerotic disease. As with examinations in which there is no aortic enhancement at all, aortic dilatation, contour defects, and findings suggestive of rupture can increase confidence that a not yet filling false lumen is present. Particularly on a single-phase examination, a nonfilling false lumen can also lead to confusion in distinguishing class I from class II lesions. In almost all of these cases, determining whether a lesion is class I with a nonfilling false lumen classic AD or class II with an IMH is not of immediate clinical importance. Thus, in reporting these findings, we describe the appearance and communicate them as *aortic dissection equivalents*; further characterization can then be performed with repeat CT in a later phase, MRI, or TEE as needed (Fig 7). Finally, a delayed-filling false lumen can be confused for a thrombosed false lumen in the early systemic arterial phase. Therefore, recognizing the early systemic arterial phase will prevent confusion with a thrombosed false lumen; this distinction is important as a thrombosed false lumen has been associated with improved outcomes in class I lesions (35) (Figs 8, E3).

Unrecognized Pitfall: Flow Artifacts

As experience with CT for AAS has increased, radiologists have identified important pitfalls in interpretation. Many, such as pulsation artifact in the ascending aorta at non-EKG gated examinations, are commonly encountered and have been mitigated with improvements in modern CT scanner technology including EKG-gating and high-pitch imaging modes (12). Faster CT scanners, however, may be responsible for an increased risk of flow artifacts, which are best described as nonuniform

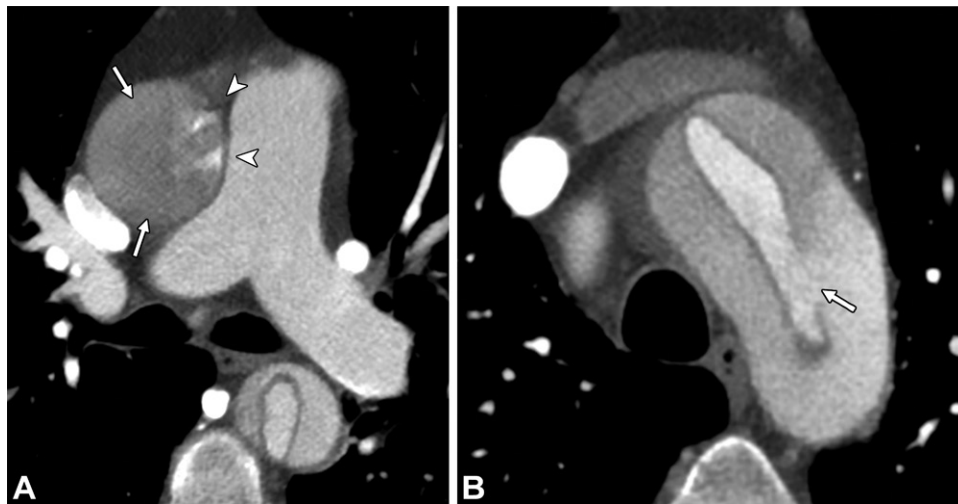


Figure 6. Stanford type A and SVS type B dissection in a 60-year-old man who presented with chest pain. **(A)** Axial CT image obtained in an early systemic arterial phase shows an intimomedial flap in the ascending and descending aorta, with nonenhancing false lumen (arrows) and severely compressed true lumen (arrowheads) in the ascending aorta and intimomedial flap with enhancing true and false lumens in the descending aorta. **(B)** Axial CT image obtained at the level of the aortic arch shows the entry tear (arrow) at the point where the contrast material opacification equilibrates between the true and false lumens. This is a Stanford type A dissection but, according to the new SVS-suggested classification system, it is a type B dissection, as the entry tear is outside of zone 0.



Figure 7. Type A AD equivalent in a 51-year-old man who presented with chest pain. **(A)** Axial CT image obtained from a PE protocol with transient interruption of the contrast column shows contrast material in the aorta but no contrast material in the pulmonary arterial tree. Crescentic fluid is present about the ascending aorta (solid arrow), with hemopericardium (arrowheads) and blood tracking around the right pulmonary artery (dashed arrows) consistent with shared sheath rupture. **(B)** Axial CT image obtained more superiorly than **A** shows additional fluid (arrow) tracking around the aortic arch. In this early systemic arterial phase, it is difficult to determine whether the crescentic fluid is a nonfilling false lumen, thrombosed false lumen, or IMH. In this setting, we refer to this lesion as an AD equivalent. At surgery, this lesion was found to be a classic AD with a patent false lumen.

opacification of the blood pool at postcontrast examinations (36). While flow artifacts have been reported to be the cause of 11.1% of PE misdiagnosis cases in one series, their impact on the interpretation of examinations performed for AAS is not known (37). Our practical experience has indicated that flow artifacts are an unrecognized pitfall and cause of overdiagnosis of AD.

At CT, aortic flow artifacts can result in apparent linear filling defects in the contrast column that can simulate an intimomedial flap. The causes of flow artifacts are not always known, but they may be exaggerated by faster modern scanners and in patients with heart failure owing to the scan acquisition outracing or occurring at the leading edge of the contrast column where mixing of



Figure 8. Type B AD at PE protocol CT in a 62-year-old man who presented with chest pain. Axial CT image obtained by using a PE protocol shows a crescentic false lumen in the descending aorta. This was originally interpreted as an AD with a thrombosed false lumen. A subsequent four-dimensional flow MR angiographic GIF (Fig E3) showed that the false lumen was patent. In this early systemic arterial phase of contrast enhancement, there should be caution in interpreting the false lumen as thrombosed.

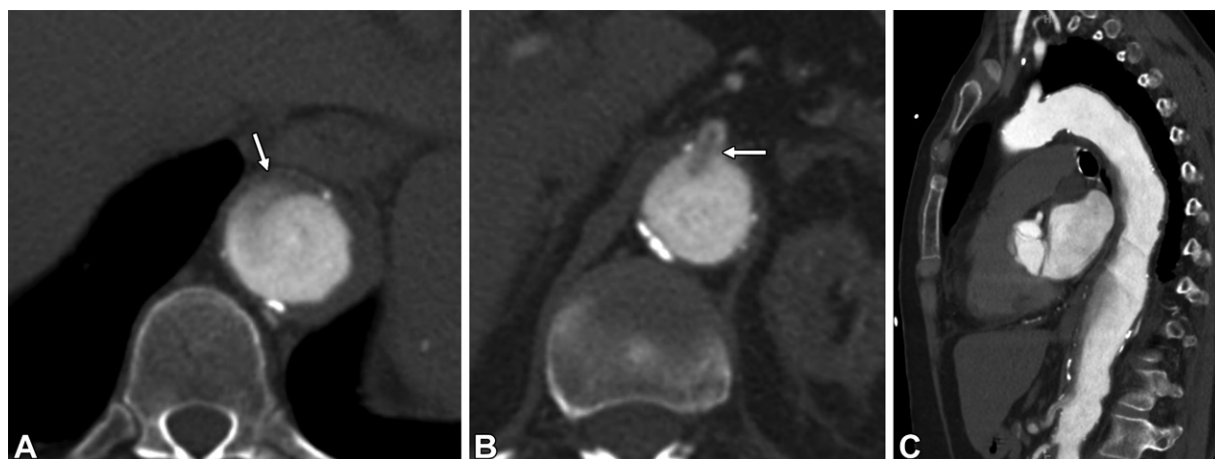


Figure 9. Flow artifact extending into the celiac axis at CT in a 70-year-old woman who presented with chest pain. (A, B) Axial CT images show a filling defect along the anterior aspect of the aorta with indistinct borders that extends into the celiac axis (arrow). (C) Oblique sagittal reconstruction through the filling defect confirms that it has an indistinct "smoke-like" appearance. These findings are typical of a flow artifact. TEE (not shown) helped confirm that there was no intimal flap in the aorta.

opacified and nonopacified blood is more likely. In the aorta, flow artifacts are more common along the lesser curvature of the arch and along the central aspect of the descending aorta, potentially owing to slow and/or turbulent vascular flow.

There are several features of flow artifacts that can assist in distinguishing them from a true intimal flap. Given that they are caused by mixing of opacified and nonopacified blood, flow artifacts typically have higher attenuation than an intimal flap (38). More importantly, flow artifacts often have indistinct borders and discontinuities in at least one plane and have been described as "smoke-like" (Fig 9). The indistinct borders of flow artifacts are better appreciated on reconstructed views along the long axis of the vessel. Thus when suspected, multiplanar reconstructions are often of value. Finally, flow artifacts may not follow the expected anatomic

distribution of a classic AD as they often begin more inferiorly in the descending aorta than typical true intimal flaps.

When a flow artifact is suspected but equivocal on the basis of its CT appearance, several options can be pursued to better characterize it. The first option is repeating the CT examination with a longer scan delay from contrast material injection to mitigate the flow-related effects of the leading edge of the contrast column (Fig 10). Of note, although rare, flow artifacts visible at CT can reproduce when rescanning patients (Fig 11). Therefore, if the suspicion that a finding is still a flow artifact persists at the repeat examination, other imaging modalities will need to be utilized. One option is the use of noncontrast balanced steady-state free-precession MRI, which can be performed with prospectively gated single-shot and retrospectively gated cine technique through

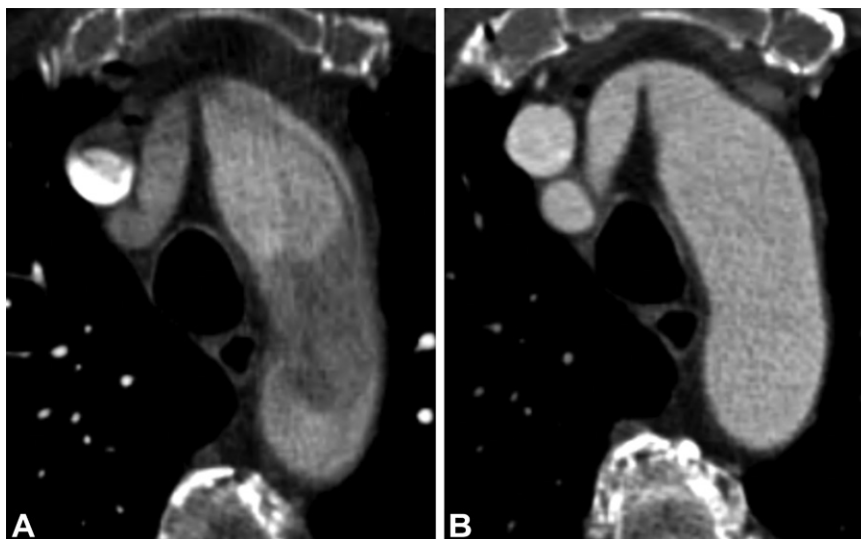
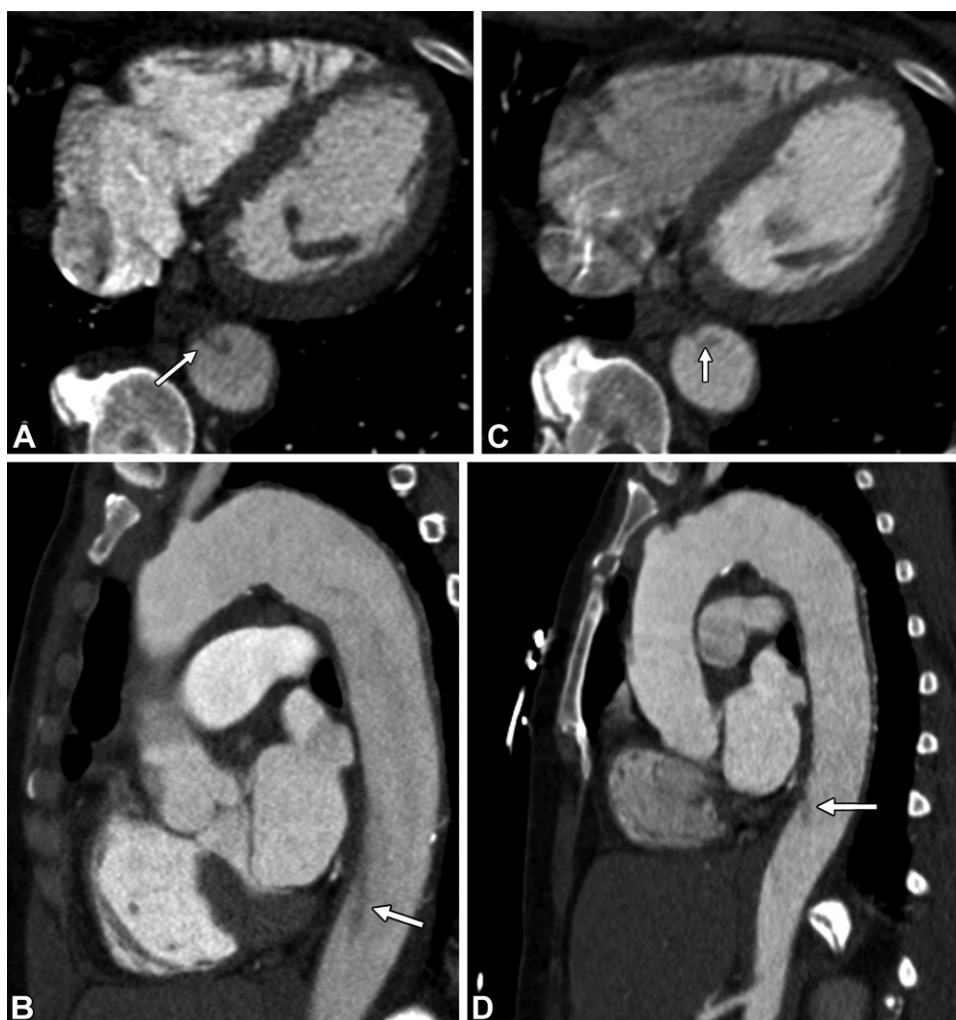


Figure 10. Flow artifact in the aortic arch and proximal descending thoracic aorta in an 85-year-old woman with a history of decreased ejection fraction who presented with chest pain and underwent dissection protocol CT. (A) Initial acquisition axial CT image shows an unusual filling defect in the aortic arch that has a somewhat linear character that mimics an intimal flap. This was identified at the CT scanner, and the patient was reimaged in the venous phase. (B) The filling defect depicted in the aortic arch on the initial image is not present on the venous phase axial CT image, confirming that it is a flow artifact.

Figure 11. Reproducible flow artifact in the descending aorta at CT in a 73-year-old woman who presented with chest pain. (A, B) Axial (A) and oblique sagittal (B) CT images show a linear filling defect (arrow) that was indistinct on axial images and “smoke-like” on the oblique sagittal reconstruction. This was favored to be a flow artifact. (C, D) Repeat CT was performed, and the axial (C) and oblique sagittal (D) images show a similar linear filling defect (arrow) in the descending thoracic aorta. At this point, a flow artifact was still suspected and TEE was performed, which confirmed that no dissection was present (not shown).



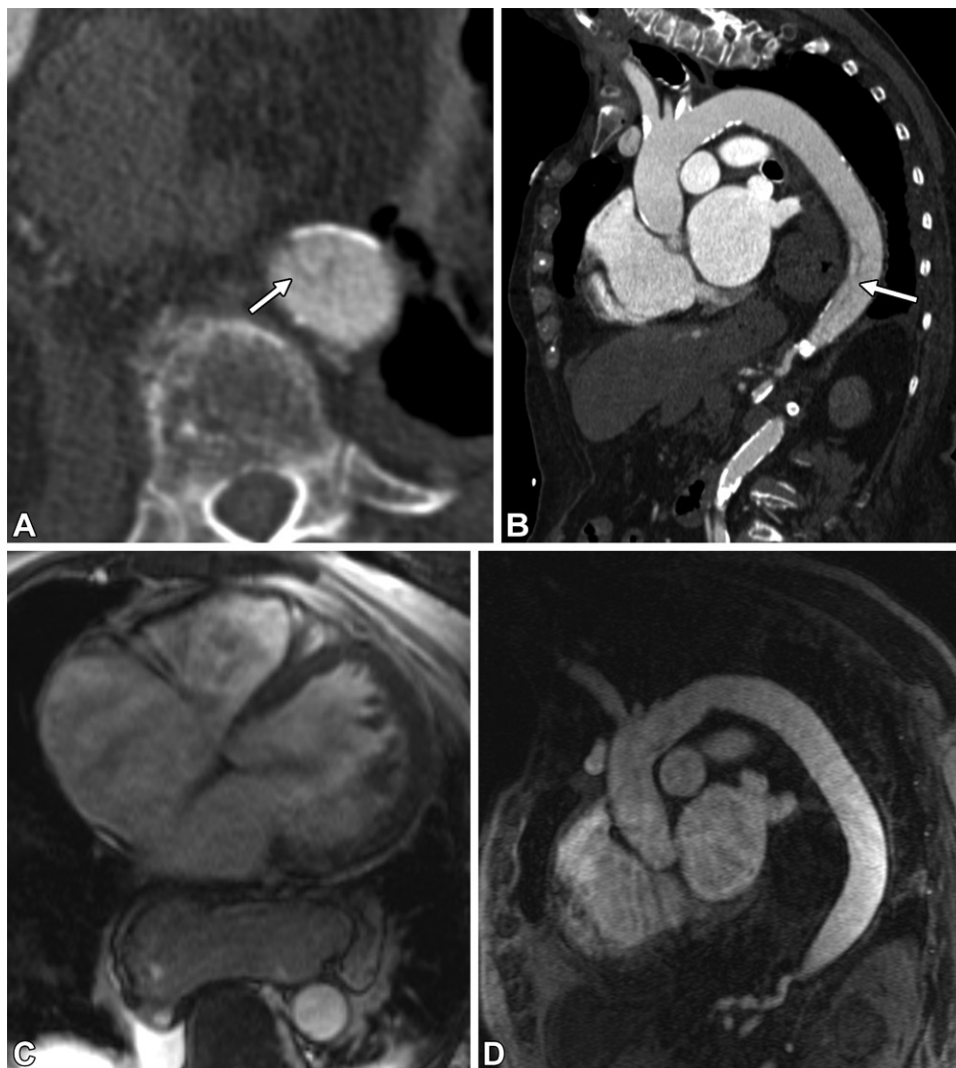


Figure 12. CT flow artifact in the descending aorta confirmed at MR angiography in a 74-year-old woman with chest pain who underwent dissection protocol CT. (A, B) Axial (A) and oblique sagittal (B) CT images show a linear filling defect (arrow) in the descending aorta that was originally interpreted as an AD, prompting transfer to our institution. A rereview of this finding raised suspicion that it was a flow artifact, given that it was thin and somewhat indistinct, was discontinuous, and originated more distally than a typical type B AD. (C, D) Axial balanced steady-state free-precession MR image (C) and oblique sagittal postcontrast MR angiographic image (D) do not redemonstrate the linear filling defect, confirming that it was a flow artifact.

the region of interest to confirm or refute the presence of an intimomedial flap (Fig 12). Another is performing TEE in the region of interest.

Difficulties in AAS Classification: More than Type A/B—Terminology and Communication

The Stanford classification system is most often used by radiologists when communicating the anatomic extent of acute aortic pathologic conditions. Pathologic conditions that involve the ascending aorta are classified as type A, while those involving the descending aorta distal to the left subclavian artery are type B (10). This classification system is based on the general principle that type A conditions are managed surgically

while uncomplicated type B conditions can be managed medically. Dissections that begin in the aortic arch and extend distally are not explicitly described in the original Stanford classification system and have been the subject of some controversy, particularly in the last decade. Given that the majority of uncomplicated dissections beginning in the aortic arch and extending distally are managed medically, these are best classified as type B ADs with arch involvement, as has been described by Lempel et al (39).

With the expanded use of endovascular techniques in aortic repair, the Society for Vascular Surgery (SVS) has introduced new guidelines for the description of acute aortic pathologic conditions. While the A/B dichotomy of the Stanford

system is still used, a more detailed anatomic description of the extent of acute aortic pathologic conditions using the aortic zones as originally described by Criado and Ishimaru (40) is encouraged (Fig E4). Importantly, the location of the entry tear (which is the most proximal discontinuity in the intimomedial flap in class I lesions) rather than the extent of the intimomedial flap is used to determine whether a classic AD is type A or type B (40) (Fig 6). When the entry tear is not identifiable, classification as type I (indeterminate) is suggested, which can later be revised if an entry tear is discovered at subsequent imaging. This change is relevant to therapy, as sealing the entry tear is the primary target of interventions in patients with class I lesions. As such, an AD with an entry tear in zone 3 and a false lumen spanning zone 0 to zone 5 would be described as SVS type B_{0,5} (40).

The SVS suggests IMH is classified similarly and based only on the longitudinal extent of the IMH, with additional notation given to the presence of associated PAU. The SVS guidelines do make reference to ulcerlike projections (ULPs), a term that has been used to describe intimomedial defects in the setting of IMH, but describe the terms as interchangeable (40). This is a reflection of the terminology used to describe intimomedial defects in the setting of IMH, in which PAU, ULP, and other terms, including focal intimal disruption (FID) and tiny intimal disruption (TID), are used with attempted but inconsistent definitions among different authors in the literature and in communication between physicians (8,13,17,41,42).

While the terminology is inconsistent, intimomedial defects in the setting of IMH with larger communications with the aortic lumen have been shown to have a negative association with patient outcomes. In some cases, they can change over serial examinations and appear to function similar to the entry tear in a classic AD, with their interval development and enlargement being associated with increased propagation or the development of complications (13,17,43,44). As such, they may need to be a focus of treatment in IMH even if they are not located within the thickest part of the IMH or are located outside zone 0 in a type A lesion. This is in contrast to focal contrast material collections in the wall of an IMH that are isolated or have a small vessel traversing them, so-called intramural blood pools, which are thought to be due to avulsed branch vessels and of no prognostic significance (13,45) (Figs 13, E5).

It is interesting to note that per the SVS writing committee, the new proposed classification system “will allow clinicians to conceptually project a precise image of entry tear location and proximal and distal dissection extent with one

simple designation,” but that it is “not necessarily the intent” of this system to “completely replace the current Stanford and DeBakey systems for everyday clinical use, especially for practitioners who are not subject matter experts in aortic disease.” These statements accurately reflect the difficulty in implementing new and subtly changing terminology in cases of acute aortic pathologic conditions, the treatment of which is continually evolving often on the basis of personal preferences and experience. Diagnostic radiologists must acknowledge that cases of acute aortic pathologic conditions often require direct communication with treating physicians to ensure that what is being depicted on images is accurately conveyed. This is particularly relevant given that the language used to describe these lesions is not universally agreed upon or used. This is also of importance in describing some of the difficult cases discussed previously, including lesions at CT that cannot be definitively characterized as class I classic AD or class II IMH owing to phase of contrast enhancement (which we term *AD equivalents*) or the less common class III lesions, which often have an imaging appearance that may belie their serious nature and longitudinal extent.

The Unusual Rupture and Pitfalls

In the setting of an acute aortic pathologic condition, aortic rupture is a potential complication that may lead to more urgent invasive treatment. There are several pathways in which the aorta can rupture. The most commonly seen pathway is rupture through the adventitia into the surrounding mediastinum or abdomen. CT findings of rupture through the adventitia include the presence of hematoma (intermediate- to high-attenuating material measuring 40–75 HU) adjacent to an acute aortic pathologic lesion. In some cases, active extravasation of contrast material may be visible into the hematoma, a finding which must be identified as it typically requires immediate management, while findings of rupture without active extravasation can be managed in a less urgent fashion in some stable patients (46) (Fig 14).

The aorta can also rupture into the pericardial and pleural spaces. At CT, the presence of high-attenuating blood products in these spaces, particularly when mixed with lower-attenuating unclotted blood or associated with a hematocrit sign, is diagnostic of rupture (47). In the case of rupture into the pericardium, cardiac tamponade can also occur and may require rapid decompression. CT findings of cardiac tamponade include compression of the right heart, deviation of the interventricular septum to the left, and the distention of collateral pathways with reflux of contrast material (48) (Fig 15). In a patient with

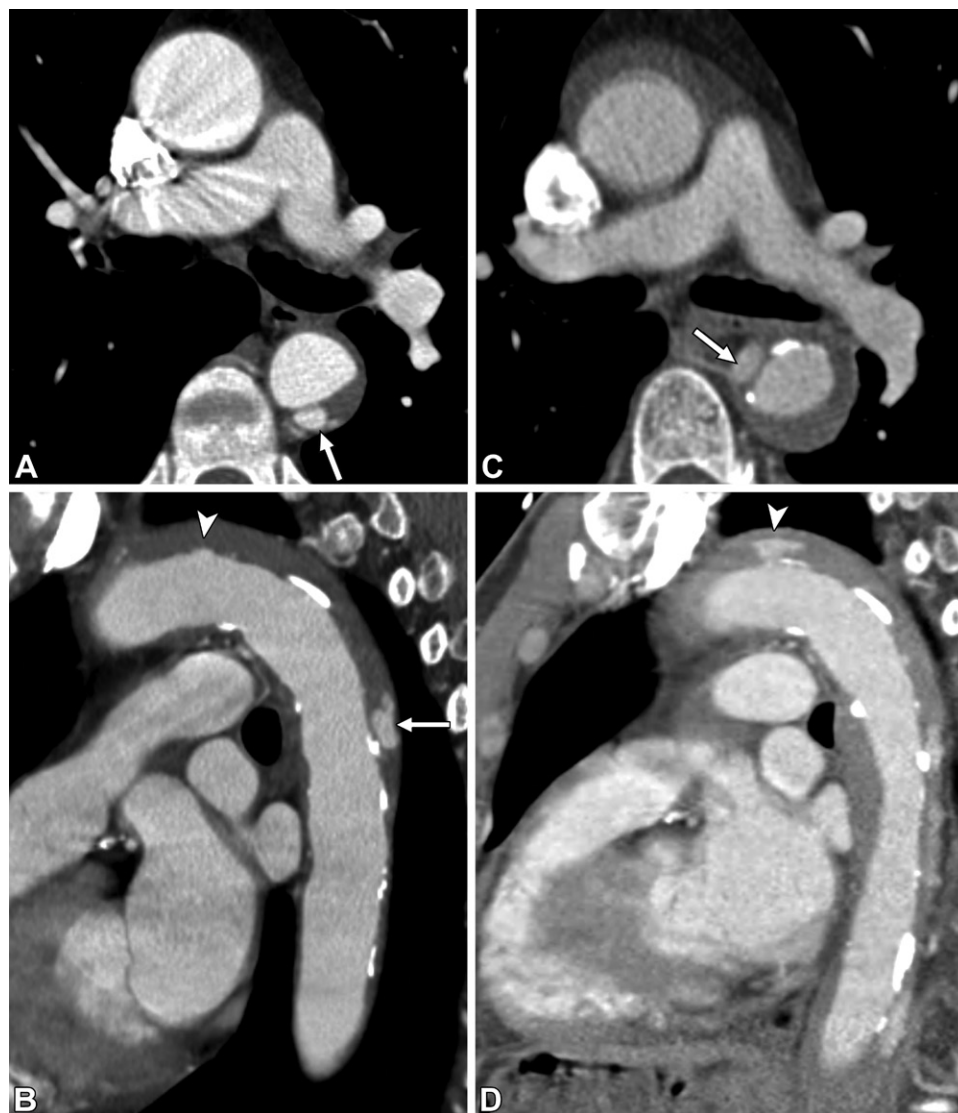


Figure 13. Type B IMH (Svensson class II lesion) with evolving ULP and retrograde conversion in a 66-year-old man who presented with chest pain. (A, B) Axial (A) and oblique sagittal (B) CT images show crescentic thickening consistent with IMH involving the descending aorta. In the proximal descending aorta, there was a small broad-based defect (arrowhead in B) in the atherosclerotic plaque that contacted the IMH; a potential ULP was questioned at this site. More distally, there were contrast material collections in the wall of the hematoma that did not communicate with the lumen, consistent with intramural blood pools (arrow). This Stanford type B IMH was managed medically. (C, D) Two days later, the patient had worsening chest pain. Axial CT image (C) shows retrograde type A extension of the IMH into the ascending aorta. On the oblique sagittal CT image (D), the suspected site of the ULP shows contrast material extension (arrowhead) into the IMH. The IMH also expanded in the descending aorta, with a new intramural blood pool depicted medially at the level of an avulsed intercostal artery (arrow in C). At this point, an endovascular stent was used to treat the ULP. The ascending aorta was managed conservatively.

chest pain, the presence of a pericardial fluid collection should trigger a close inspection of the ascending aorta as a pericardial hematoma may be the most easily identified sign of a subtle aortic pathologic condition (46).

Pericardial rupture should only occur with acute aortic pathologic conditions that involve the ascending aorta, as it is invested in the pericardium to the level of the innominate artery. As such, the presence of hemopericardium in a patient with an acute aortic pathologic condition

is indicative of extension to the ascending aorta. However, caution is indicated as there are several pitfalls in the detection of hemopericardium. First, in aortic pathologic conditions involving the arch and descending aorta, rupture of blood into the mediastinum can layer on top of the pericardium. On axial images, this can appear to involve the pericardium and be misdiagnosed as hemopericardium. In these cases, multiplanar reconstructions can be helpful in distinguishing these conditions, as mediastinal rupture layering on the pericardium

Figure 14. Rupturing Stanford type B AD in a 65-year-old man who presented with chest pain. Axial CT image shows Stanford type B dissection with hematoma surrounding the aorta and a left hemothorax. A focus of active extravasation of contrast material (arrow) is depicted along the lateral aspect of the descending thoracic aorta. These findings prompted urgent endovascular stent repair of the Stanford type B dissection.

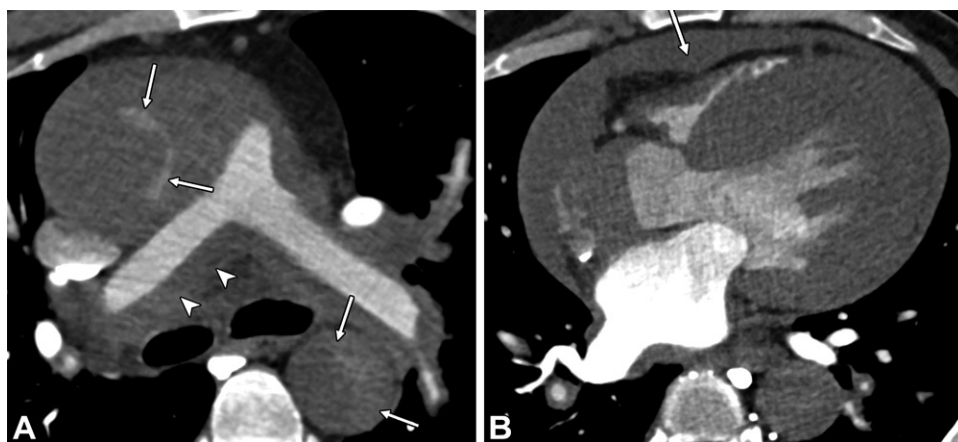
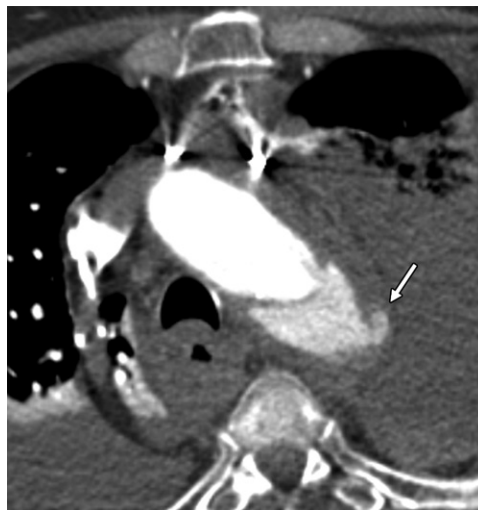


Figure 15. Ruptured Stanford type A AD equivalent with tamponade in a 48-year-old man who presented with acute chest pain and myocardial infarction. (A) Axial CT image shows very weakly enhancing true lumen in the ascending and descending aorta (arrows) in this patient with a type A AD equivalent. There are findings of shared sheath rupture with blood extending along the central pulmonary arteries (arrowheads). (B) More caudal axial CT image shows hemopericardium with compression of the right ventricle (arrow) and near obliteration of the right atrium. Note the blood extending out along the pulmonary arterial tree from the shared sheath rupture (arrowheads in A). Emergent pericardial window CT findings confirmed hemopericardium, but the patient did not survive for surgical repair.

will not be visible inferiorly and will often deviate laterally as it layers (Fig 16).

A potential pitfall in the detection of pericardial and pleural rupture is third spacing of contrast material from previous CT examinations. In patients who have undergone recent CT examinations, contrast material can accumulate in nonbloody pleural and pericardial effusions and simulate the presence of blood products. This may be a particular problem given that patients may require short-term repeat examinations to better characterize equivocal findings or to follow progression. Third spacing of contrast material can be differentiated on CT images as it will typically be more homogeneously hyperattenuating than blood products (Fig 17). In addition, when third spacing is present, it may be seen in multiple compartments (49).

Given that the aortic root has a shared adventitial sheath with the main pulmonary artery, proximal acute aortic pathologic conditions that have ruptured into the subadventitial space can extend about the pulmonary arterial tree (50,51). This was originally thought to be a rare pathway of rupture, but increased awareness and improvements in CT scanner technology indicate that it is probably more common in dissections involving the ascending aorta than originally believed. Shared sheath rupture can have different imaging appearances on CT images. In dramatic cases, a dilated ascending aorta with medial disruption can result in the extension of a large amount of blood around the proximal pulmonary arteries (8). In severe cases, this can result in acute right heart obstruction similar to that encountered in PE. The central pulmo-

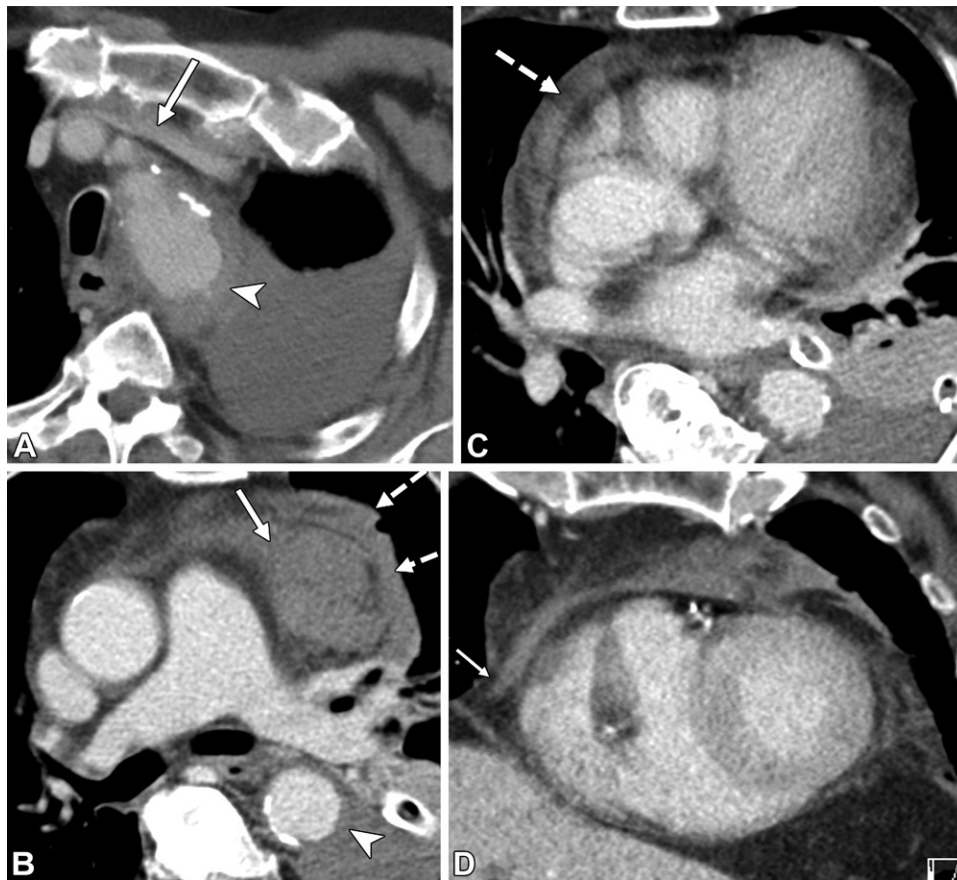


Figure 16. Mediastinal blood simulating hemopericardium in a 55-year-old man with ruptured IMH. (A–C) Axial CT images show Stanford type B IMH (arrowhead in A and B) with blood in the mediastinum (solid arrow in A and B). Inferiorly on the axial images, crescentic blood is depicted anteriorly, which was interpreted as concerning for hemopericardium (dashed arrows in B and C). (D) Coronal CT reconstruction shows that this blood is layering on top of the pericardium and deviates laterally on the right (arrow). There is no blood inferiorly within the pericardium. Echocardiographic images did not show pericardial fluid.

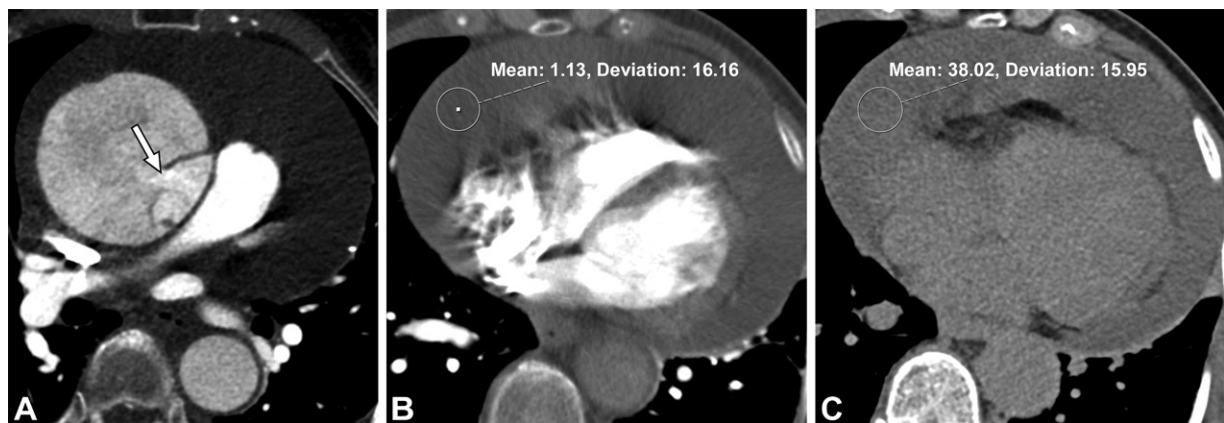


Figure 17. Third spacing in a pericardial effusion in a 60-year-old woman with a history of lupus who presented with chest pain. (A) Axial CT image shows a Stanford type A AD with contrast material extending across the entry tear (arrow) in the dilated ascending aorta. (B) This was initially identified at an outside hospital by using PE protocol CT, where a large simple-appearing pericardial effusion was depicted on the axial CT image. (C) Repeat axial CT image obtained 3 hours later shows that this pericardial effusion has increased attenuation on the noncontrast portion of the examination. This is consistent with third spacing of contrast material into the lupus-related pericardial effusion. At surgical repair, the pericardial effusion was found to be serous and nonbloody.

nary arteries may be severely compressed or even completely obstructed in association with a dilated right heart and interventricular septal

deviation to the left. While findings in severe shared sheath rupture are unlikely to be missed, they may be misinterpreted as PE and can thus

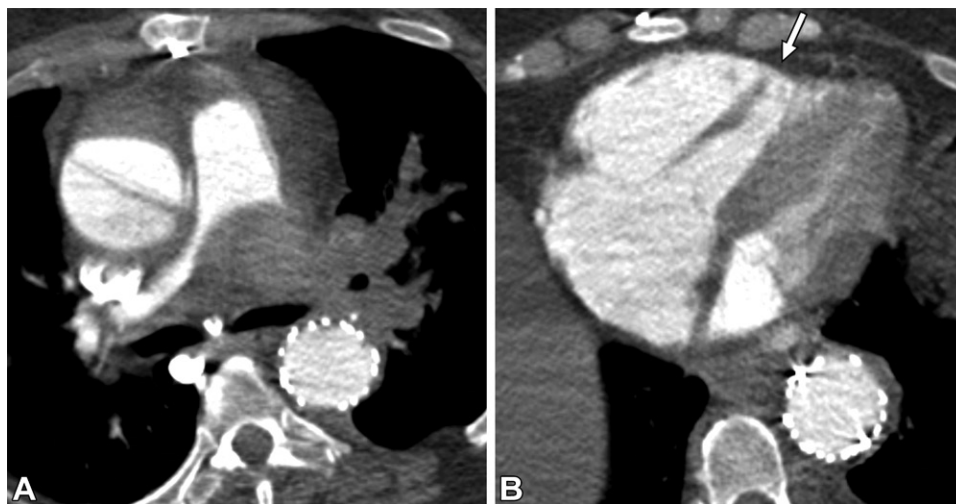


Figure 18. Stanford type A AD with shared sheath rupture and acute pulmonary hypertension in a 65-year-old woman who transferred to our institution with a diagnosis of “AD and PE.” (A) Axial CT image shows an intimomedial flap in the ascending aorta. There is complete occlusion of the left pulmonary arterial tree with relatively high-attenuating material. In the setting of the AD, this appearance is consistent with shared sheath rupture, not PE. (B) Axial CT image of the heart shows marked right heart enlargement at the base and midcavity with notched-like contraction distally (arrow), the CT equivalent of the McConnell sign. These findings were consistent with acute pulmonary hypertension that was confirmed at echocardiography. At surgery, the finding of shared sheath rupture was confirmed, and the clot was evacuated from the shared sheath about the left pulmonary arterial tree.

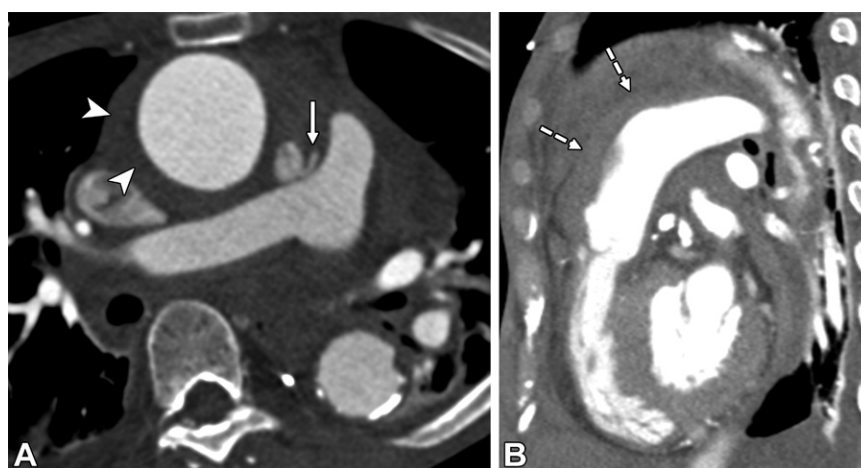


Figure 19. Shared sheath rupture with active contrast material communication in a 63-year-old woman with a Stanford type A AD equivalent. Axial (A) and oblique sagittal (B) CT images show crescentic thickening of the ascending aorta (arrowheads in A) consistent with AD equivalent. There is also active extravasation of contrast material from the proximal aorta to the main pulmonary artery via the shared adventitial sheath (arrow in A). On the oblique sagittal image, findings of shared sheath rupture extend along the main pulmonary artery (dashed arrows in B). These findings were confirmed at surgical repair.

lead to potentially fatal consequences due to unnecessary anticoagulation therapy or thrombolysis. Identification of severe compression of the pulmonary arterial tree may also indicate the need for the treating surgeon to relieve this obstruction in the operating room (Fig 18).

Shared sheath rupture can also occur with smaller communications between the proximal aorta and pulmonary artery. Rarely, a direct communication between the aortic wall and the pulmonary artery may be depicted on CT

images (Fig 19). More commonly, no direct communication is identified but hematoma will be depicted eccentrically about the wall of the proximal pulmonary arteries. This hematoma can track into the branch pulmonary arteries and has been reported to be associated with ground-glass findings of parenchymal hemorrhage. Again, recognizing these less severe forms of shared sheath rupture can prevent misdiagnosis of PE. More importantly, however, the identification of shared sheath rupture can be instrumen-

tal in directing the radiologist to carefully assess for subtle pathologic findings in the aortic root.

Conclusion

As CT has emerged as the first-line test in the assessment of patients with suspected acute aortic pathologic conditions, interpretation of these examinations has become a necessary component of general radiology practice. Fortunately, in the majority of cases, the basics of interpretation are not difficult and have been extensively reviewed over the past several decades. This article highlights several important concepts that extend this previous work with the goal of identifying nuances and pitfalls that can aid in the interpretation of difficult cases and findings. We describe issues pertaining to diagnosis, including recognizing the Svensson classes of intimo-medial tears, findings of flapless AD equivalents, pitfalls with phase of contrast enhancement, and flow artifacts. We discuss the classification of acute aortic pathologic conditions, including emerging recommendations from the SVS. Finally, we present examples and pitfalls involving the assessment of aortic rupture. These are all concepts that we have encountered in difficult cases of acute aortic pathologic conditions. Ultimately, when used in conjunction with more widely available material on the acute aorta, these concepts can be used to add nuance and value to CT interpretation in this important patient cohort.

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